Telecommunication Assignment -7

Faculty of Engineering

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Technology,Business and Design

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Task - 1

Simulate the Angle-of-Arrival (AOA) coverage of for antenna array arrangements.

1) Define an equilateral antenna array (triangle with equal distances/length) for 3 antennas with

a) 0.5 lambda wavelength (WL)

b) 1.0 lambda wavelength (WL)

c) 1.5 lambda wavelength (WL)

2) Create synthetic RANDOM complex data to calculate AOA of suitable size

   Plot/show example I/Q-diagram of the data and histograms of I/Q data

3) Derive the AOA for the synthetic data, for a,b,c configurations

   Plot the AOA in dcosx,dcosy

Script :

```
import numpy as np
import matplotlib.pyplot as plt

def antenna_positions(wavelength):

    d = wavelength / 2
    positions = np.array([[0, 0], [d, np.sqrt(3) * d], [2 * d, 0]])
    return positions

def generate_synthetic_data(size):

    data = np.random.normal(size=(size,)) + 1j * np.random.normal(size=(size,))
    return data

def calculate_angle_of_arrival(data, antenna_positions):

    aoas = []

    for pos in antenna_positions:
        steering_vector = np.exp(-1j * 2 * np.pi * pos[0] * np.sin(np.arange(0, np.pi, 0.01)) + 1j * 2 * np.pi * pos[1] * np.cos(np.arange(0, np.pi, 0.01)))
        correlation = np.abs(np.dot(data.reshape(-1, 1), steering_vector.reshape(1, -1))) ** 2
        aoa = np.argmax(correlation, axis=1)
```

```
        aoas.append(aoa)

    return np.array(aoas).flatten()
wavelengths = [0.5, 1.0, 1.5]
data_size = 1000
arrays = []
aoas = []

# Generate antenna arrays and calculate AOAs for each wavelength
for wl in wavelengths:
    positions = antenna_positions(wl)
    arrays.append(positions)
    data = generate_synthetic_data(data_size)
    aoa = calculate_angle_of_arrival(data, positions)
    aoas.append(aoa)

  # Plot the AOA coverage
    plt.plot(aoa_range, aoa_coverage, label=f'{array_length} lambda')

# Plot settings
plt.xlabel('Angle-of-Arrival (degrees)')
plt.ylabel('Magnitude')
plt.title('Angle-of-Arrival (AOA) Coverage')
plt.legend()
plt.grid(True)
plt.show()

# Plot I/Q diagram

data = generate_synthetic_data(data_size)
plt.figure(figsize=(6, 6))
plt.scatter(data.real, data.imag, s=5)
plt.xlabel("I")
plt.ylabel("Q")
plt.title("I/Q Diagram")
plt.grid(True)
```

```
plt.show()

# Plot histograms of I/Q data

plt.figure(figsize=(12, 4))
plt.subplot(1, 2, 1)
plt.hist(data.real, bins=50, density=True)
plt.xlabel("I")
plt.ylabel("Probability")
plt.title("Histogram of I")

plt.subplot(1, 2, 2)
plt.hist(data.imag, bins=50, density=True)
plt.xlabel("Q")
plt.ylabel("Probability")
plt.title("Histogram of Q")
plt.tight_layout()
plt.show()

# Plot AOA in dcosx, dcosy

plt.figure(figsize=(12, 4))
labels = ["0.5 WL", "1.0 WL", "1.5 WL"]
for i in range(len(aoas)):
    plt.subplot(1, 3, i+1)
    plt.scatter(np.cos(aoas[i]), np.sin(aoas[i]), s=5)
    plt.xlabel("dcosx")
    plt.ylabel("dcosy")
    plt.title(f"AOA - {labels[i]}")
    plt.grid(True)

plt.tight_layout()
plt.show()
```

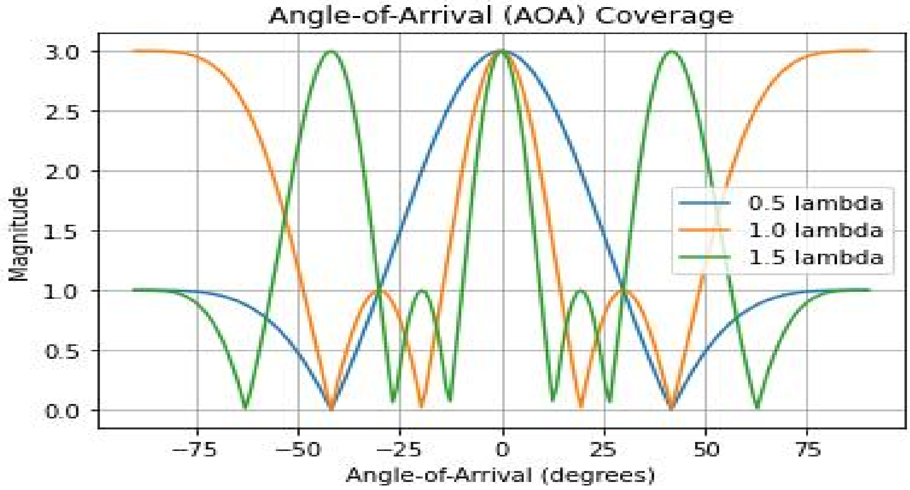

Fig 1.1 : AOA for different wavelength

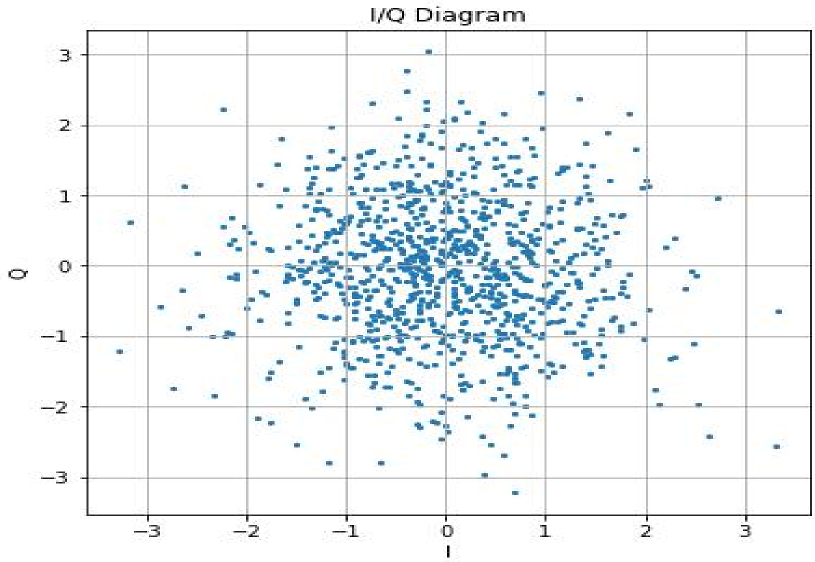

Fig 1.2 : I/Q Diagram of the Synthetic Data

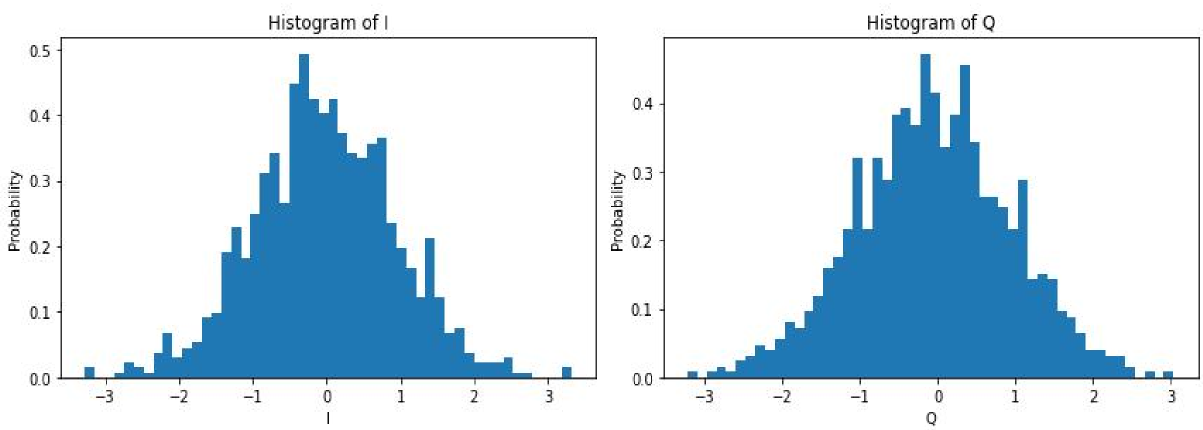

Fig 1.3 : Histogram of the I and Q data

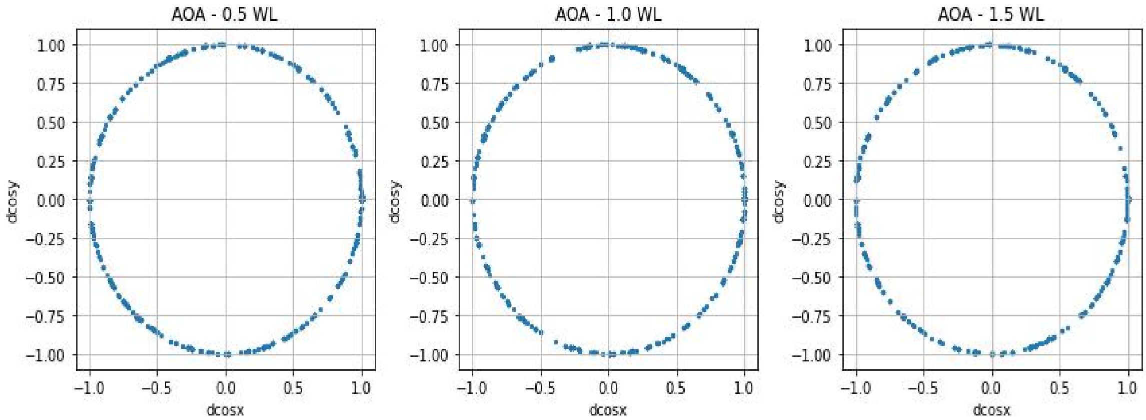

Fig 1.4 : AOA Plots in dcosx and dcosy for wavelengths 0.5, 1.0, and 1.5

References

[1] ‘Array Signal Processing ’ [online]

https://www.cs.princeton.edu/courses/archive/spring19/cos463/labs/lab5_preview.html

[accessed 07.06.2023]

[2] ‘Angle-of-Arrival’ [online]

https://www.sciencedirect.com/topics/engineering/angle-of-arrival [accessed 07.06.2023]